Yu-Jui Liang

2027 RDSS | Software / System / Validation Engineer

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Software Engineering | Software Testing & Validation | Data Center / GPU Systems | EDA / VLSI CAD

EDUCATION

National Taiwan University *Sep. 2025 – Expected Jun. 2027*

M.S. Candidate, Data Science Master Degree Program
Ongoing Research: Data Center / GPU Systems, Memory Architecture, HW–SW Co-Design, LLM Serving
Research Advisors: Prof. Jie-Hong Roland Jiang (NTU); Dr. Hsiang-Yun Cheng (Academia Sinica)

National Chengchi University *Sep. 2021 – Jun. 2025*

B.S. in Computer Science; Double Major in Electronic Physics

SOFTWARE / SYSTEMS RESEARCH & PROJECTS

Memory-Aware Hardware–Software Co-Design for Data Center LLM Serving *Oct. 2026 – Present*

Resource Scheduling / Data Movement / Design-Space Exploration

- Studying hardware–software co-design for LLM serving on heterogeneous GPUs, with prefill / decode disaggregation and hierarchical accelerator / host memory systems.
- Investigating simulation-driven scheduling, memory placement, data movement, and throughput / latency / energy trade-offs in disaggregated data centers.

SAT-LEX – LLM and SMT-Based Document Verification System *Nov. 2025 – Mar. 2026*

LLM Application / Constraint Solving / Formal Verification / Automated Reasoning

- Built an LLM-assisted document verification pipeline that translates domain rules and extracted evidence into structured constraints checked by an SMT solver.
- Detected logical conflicts, inconsistent conditions, and unsupported conclusions using deterministic solver-based checks beyond LLM-generated reasoning.

ICCAD 2025 CAD Contest Problem C – STA-Driven Incremental Timing ECO Flow

CAD Flow Automation / STA / Heuristic Search / OpenROAD

- Built an OpenROAD-based incremental ECO automation flow and designed an STA-guided branch-list heuristic to combine timing-repair operations into candidate sequences.
- Automated parasitic-aware QoR checks and rollback; earlier ECO experiments across five benchmarks improved **median setup TNS by 16.7%** and **average setup WNS by 14.6%** vs. seed DEFs.

Multi-Output Standard-Cell Mapping with MILP and Differentiable Optimization *Apr. 2026 – Present*

C++ / EDA / Technology Mapping / MILP / GPU-Accelerated Optimization

- Extended a cut-based mapper with multi-output candidate discovery, Boolean matching, and MILP-based selection; integrated candidates into differentiable optimization.
- Evaluated 30 arithmetic benchmarks with library-based area/delay and correctness checks, achieving **6% average ADP improvement** over single-output mapping.

SOFTWARE EXPERIENCE

AIMobile Co., Ltd. (Inventec Group) *Apr. 2025 – Jul. 2025*

Contract Software Engineer

- Contributed to software development, system integration, testing, and debugging in a corporate R&D environment.

Neopower Technologies Co., Ltd. *Jun. 2023 – Mar. 2025*

Contract Software Engineer – System Testing & Validation

- Developed a custom **system simulator** to reproduce operational and error scenarios, enabling repeatable software testing and validation before deployment instead of relying only on live-system testing.
- Performed end-to-end **software testing, regression validation, and post-deployment checks**, including customer-side A/B testing and log / data analysis to diagnose failures and verify system behavior after releases.

TECHNICAL SKILLS

Programming / Tools: C/C++, Python, Bash, Tcl, Linux, Git, Docker
Software Engineering: Computer Architecture, OS, System Simulation, Software Testing / Validation, Test Automation
GPU / AI Systems: CUDA, PyTorch, GPU Systems, Memory Systems, Hardware–Software Co-Design
EDA / IC Design: Logic Synthesis, Technology Mapping, MILP, STA / Timing ECO, APR, RTL-to-GDS, PPA Analysis